Polymeric Chemistry
DSC and TGA

Universität Stuttgart

Author 1:	Laurens Viehoff
Author 2:	Felix Roth
E-Mail 1:	st183688@stud.uni-stuttgart.de
E-Mail 2:	st188157@stud.uni-stuttgart.de
Student-ID 1:	3676761
Student-ID 2:	3711192

Assistentin:

Abstract

Inhaltsverzeichnis

1 Theory Part **2**

1.1 DSC 2

1.1.1 Glass transition temperature (T_g) and melting temperature (T_m) 2

1.2 TGA 3

2 Results **4**

3 Questions **5**

3.1 Question 1 5

3.2 Question 2 5

3.3 Question 3 5

3.4 Question 4 6

3.5 Question 5 6

3.6 Question 6 6

3.7 Question 7 6

3.8 Question 8 6

1 Theory Part

1.1 DSC

The DSC (Differential Scanning Calorimetry) uses the principle of the isotherm calorimetry. The Sample and a references get heated in two separate crucibles. The absorbed energy is measured, and in case of different absorption the one with less gets more power while this added power is being measured and plotted against time (t). The Integral of dW/dt is the difference between the sample and the reference, also shown in picture (1) is the Integral underneath the curve which defines the smelting enthalpy which is calculated by using equation (1). This curve is influenced by the spectrum of lengths the polymer provides, meaning if the polymer has many chains with different lengths the peak gets wider to the right. The reference is always a known Sample or something, the heat capacity is known. In our case the reference was the empty crucible.

$$\Delta H_m = \Delta H_{\text{Sample}} - \Delta H_{\text{Reference}} \tag{1}$$

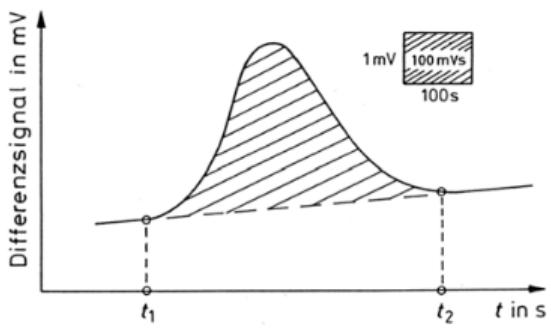


Abb. 1: This picture shows the thermogram of a Polymer, the differential signal in [mV] is plotted against time in [s]. xxx

Via this melting enthalpy the crystallization grade can be determined by using the melting enthalpy and by using ΔH_m^0. ΔH_m^0 cannot directly be determined, because polymers are not fully crystalline. Therefore, a extrapolation is used to determine the ideal case. Meaning the length of the polymer is imagined to be infinite This all results in following equation.

$$\alpha = \frac{\Delta H_m}{\Delta H_m^0} \tag{2}$$

1.1.1 Glass transition temperature (T_g) and melting temperature (T_m)

The Glass transition temperature (T_m) is the temperature range where a complete coupled segment movement of chains (10-15 chain links) is possible. These chains were frozen before that temperature and can only be defined if the polymer is amorphous. This can also be seen in a change of the heat capacity.

The melting temperature (T_g) is only present in semi crystalline polymers. These have a lamellar structure(folding crystal) with amorphous parts between the backfolding parts. When reaching

the smelting temperature these backfolded parts are moving apart. The smelting temperature can be influenced by the amount of amorphous parts in the polymer, meaning if there are more amorphous parts the smelting temperature gets higher.

Both of these temperatures can be analyzed and read out of the thermogram. For semi crystalline polymers with a bigger amorphous share the peak is more to the right because of the higher temperature.

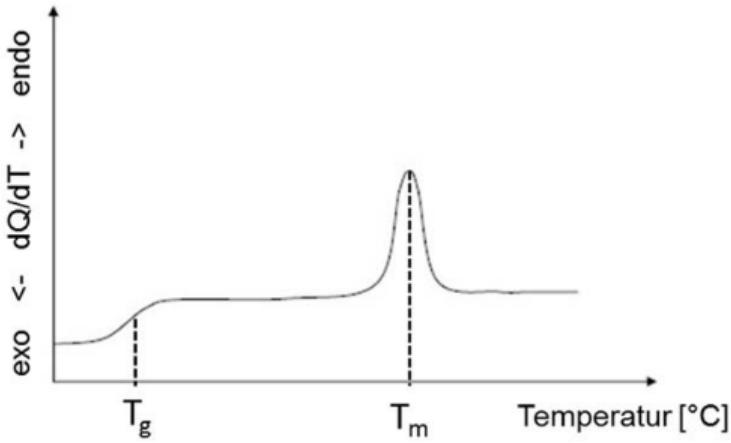


Abb. 2: This picture shows an example of a thermogram of a polymer, in which T_g is the Glass transition temperature and T_m is the melting temperature.

1.2 TGA

The main principle of TGA (Thermo gravimetry) is the heating of a Sample to the point where it gets destroyed. In the process the mass is monitored and plotted against temperature. It is also possible to plot the temperature against the time. If these two graphs are compared a clear corelation can be observed.
For analytical purposes the m-T-curve ist the more valuable one because from this curve things like the starting temperature and the end temperature of the degrading can be determined by tangent constructions. This can also be observed in picture (3).

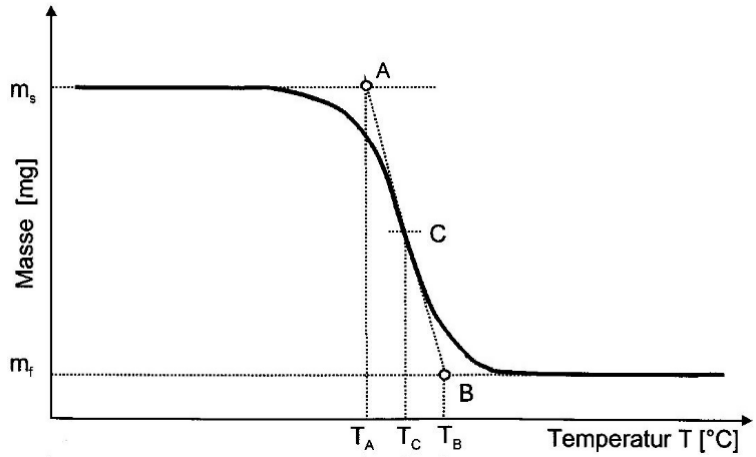


Abb. 3: The figure shows the m-T-curve for a polymer. Also shown in the picture are the points A (starting point), B (end point) and point C (middle point) and the three correlating temperatures.

Via this diagram the material specific properties can be characterized.

2 Results

Two different samples were prepared, and analyzed, using DSC and TGA. The Data of the samples (PET and *it*-PS) are listed in the Tables 1 and 2 below.

Tab. 1: Listed are the meltingtemperatures(T_m), glastemperatures(T_g) and the melting enthalpy (Δ H$_m$) of the heatingcurves if *it*-PS and PET.

Sample	Δ H$_m$ [$\frac{J}{g}$]	T$_m$ [°C]	T$_g$ [°C]
PET 1.hc	38.368	249.85	-
PET 2.hc	32.360	246.00	84.36
it-PS 1.hc	24.194	221.61	99.69

Tab. 2: Listed is the crystallizationtemperature(T_c) and the crystallization enthalpy (Δ H$_c$) of the coolingcurve of PET.

Sample	Δ H$_c$ [$\frac{J}{g}$]	T$_c$ [°C]
PET cc	-34.563	182.73

As can be seen in the attachment on pages 7 and 9, for *it*-PS were no crystallization enthalpy and no second melting enthalpy observed. Thus, before the measurement both polymers are semi-crystalline, however, the chosen cooling conditions of 20 $\frac{°C}{min}$ are too fast for the PS to crystallize again and is therefore amorphous during the measurement of the second heating curve.
A signal can also be observed in the second heating curve of the isotactic polystyrene, which occurs immediately after the glass transition temperature. This may be due to the fact that, during rapid cooling, the chains are aligned parallel to one another, as shown on the left in Figure 4, but are not crystalline.

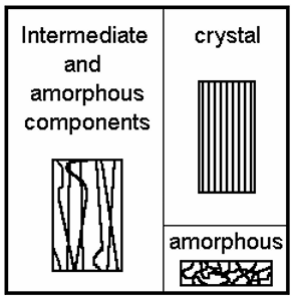


Abb. 4: The figure shows possible structural elements of the polymer chains. [1]

Upon reaching the glass transition temperature, the chains move and can align in a partially crystalline manner. This alignment triggers a melting process, which, however, occurs at low temperatures. This is because the length of the crystalline regions is very small, and according to Equation **??**, the melting temperature is very low. [1]
Using the melting enthalpy, the degree of crystallization α were calculated to

$$\alpha_{\text{PET}} = \frac{38{,}368\,\frac{\text{J}}{\text{g}}}{140\,\frac{\text{J}}{\text{g}}} = 0.274$$
$$\alpha_{it\text{-PS}} = \frac{24{,}194\,\frac{\text{J}}{\text{g}}}{87\,\frac{\text{J}}{\text{g}}} = 0.278$$

As noted previously, the *it*-PS could not crystallize under the cooling rate and therefore became amorphous. As a result, only a glass transition temperature and no melting temperature can be observed in the second heating curve.

3 Questions

3.1 Question 1

The "Big"number indicates how many monomers are needed to make one full 360° rotation. The small number in the index says how many 360° rotations are needed until the order gets repeated. For example in the case of 8_5 Polybutene. It needs 8 monomers to complete one 360° rotation and 5 full rotation until the order starts from the start.

3.2 Question 2

i-PP can be made by using the Ziegler-Natta-catalysis. In this case catalysts like $TiCl_3$ on $MgCl_2$-carrier are used in combination with aluminumalkenes. It can also be done via Metallocene-catalysis, this technique is specially used to control the tacticity and results in high graded i-PP. i-PS can also be synthesized using the Metallocene-catalysis. But can also be synthesized via Anionic-Polymerization.

3.3 Question 3

Commonly the degree of crystallization decides how stiff a polymer is. The higher the degree of crystallization, the harder and stiffer the polymer. It also gets less ductile and more dense

because crystalline parts can be packed better than amorphous parts.

3.4 Question 4

The Polymer is divided in different sections a back folding section(black), which would be the PS in our case and a amorphous section(red), which in our case would be the PE sections.

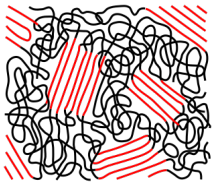


Abb. 5: The picture shows the part-.crystalline

3.5 Question 5

To get an semi-crystalline polymer out of an polymerglass it has to be reheated. By reheating the polymer the molecular mobility gets high enough to form crystals. This process is also known as cold crystallization.

3.6 Question 6

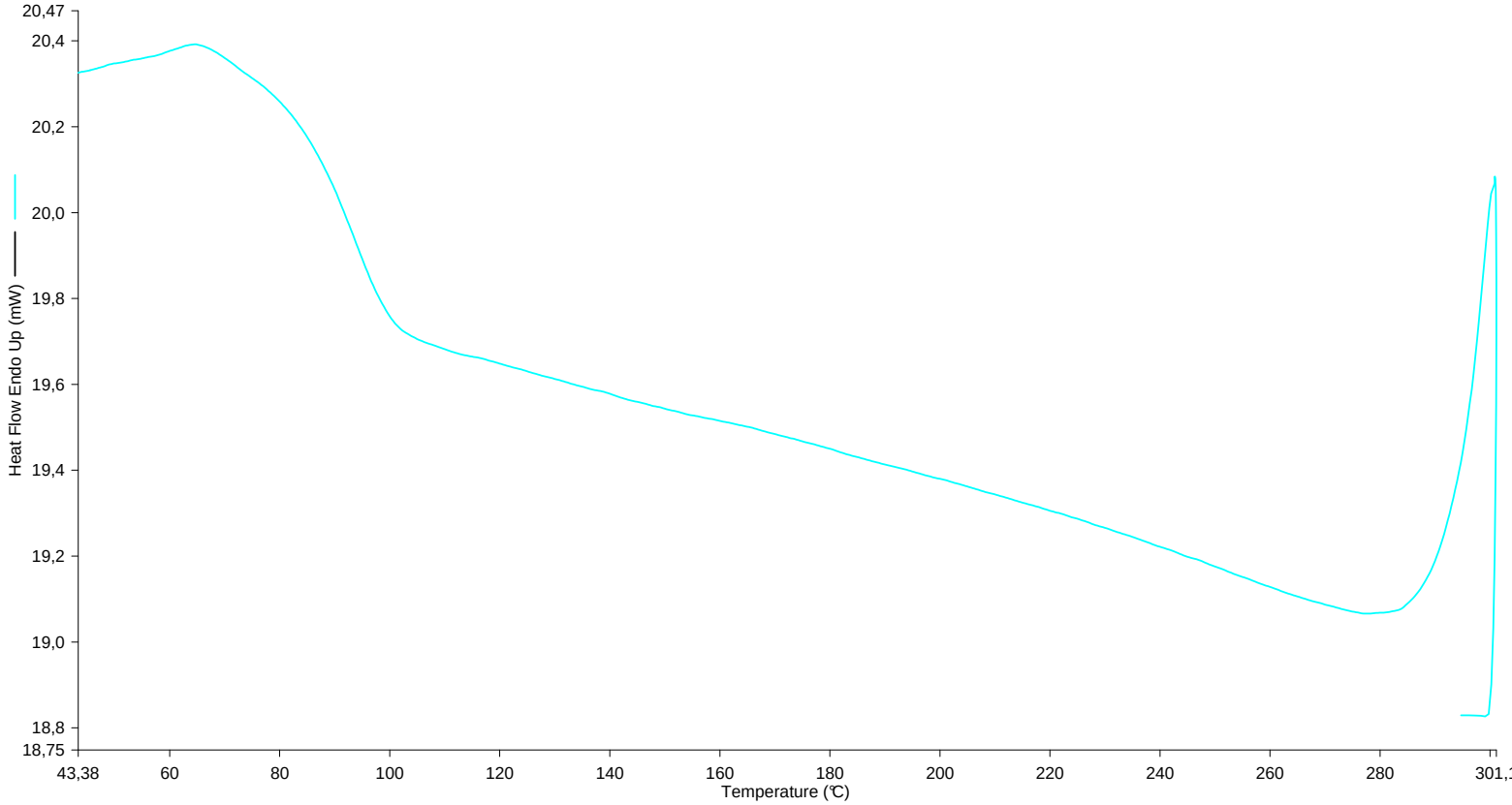
The increase in mass can be explained by oxidation. While increasing the temperature in the TGA the polymer oxidizes, the added Oxygen increases the mass of the polymer (molecule if the polymer gets destroyed).

3.7 Question 7

3.8 Question 8

Due to the residual moisture, the PET decomposes to Ethylenglycol and Terephthalic acid. The Ethylenglycol eliminates water which results in vinylalcohol, which can tautomerize to Ethanal.

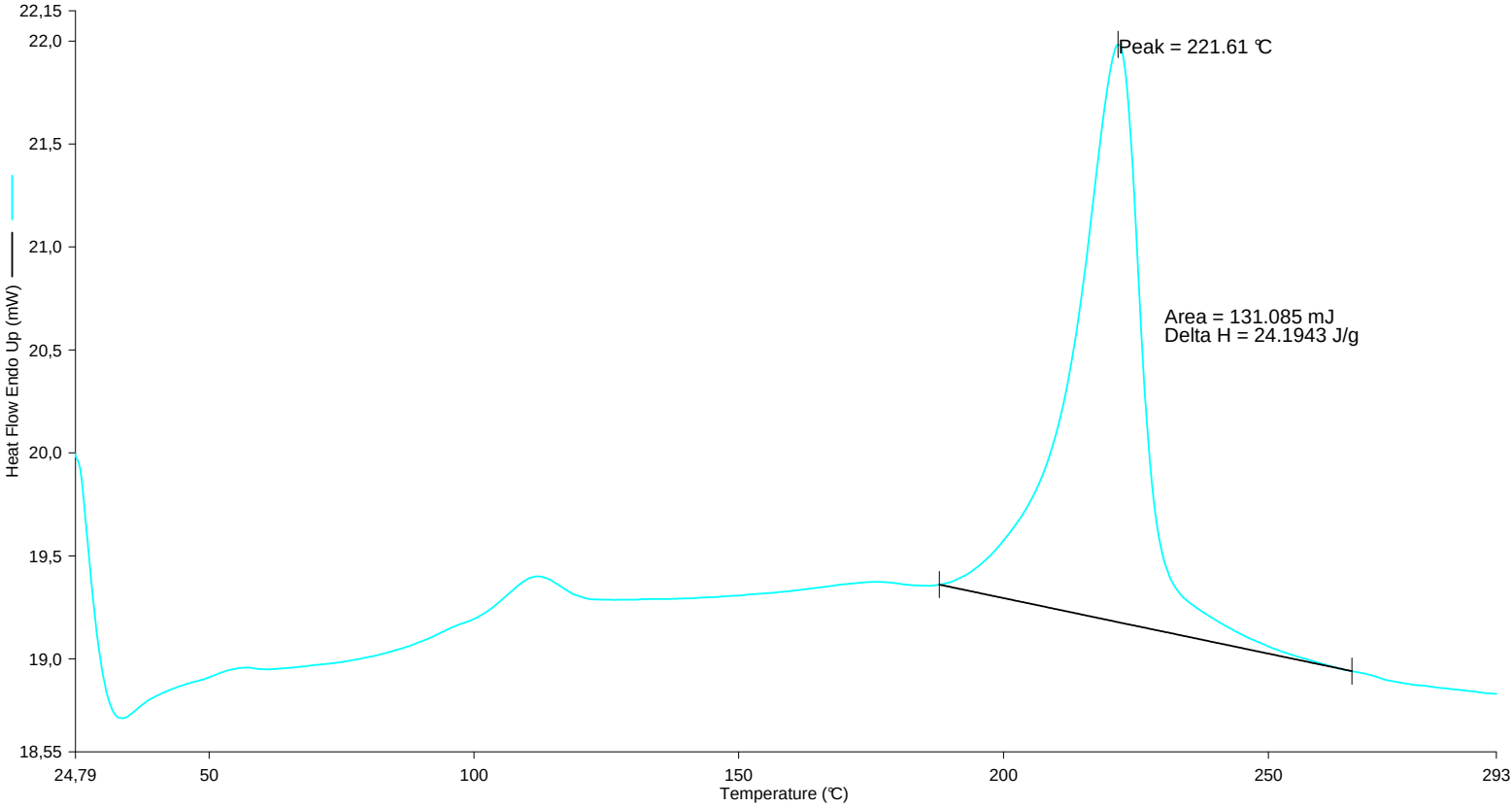
Filename:	C:\Users\DSC\Desktop\MD\G...\ps-iso-dsc.d6d
Operator ID:	MD
Sample ID:	G8-ps-iso
Sample Weight:	5.418 mg
Comment:	



10.03.2026 11:26:12

1) Heat from 30.00°C to 300.00°C at 20.00°C/min
2) Cool from 300.00°C to 30.00°C at 20.00°C/min
3) Heat from 30.00°C to 300.00°C at 20.00°C/min

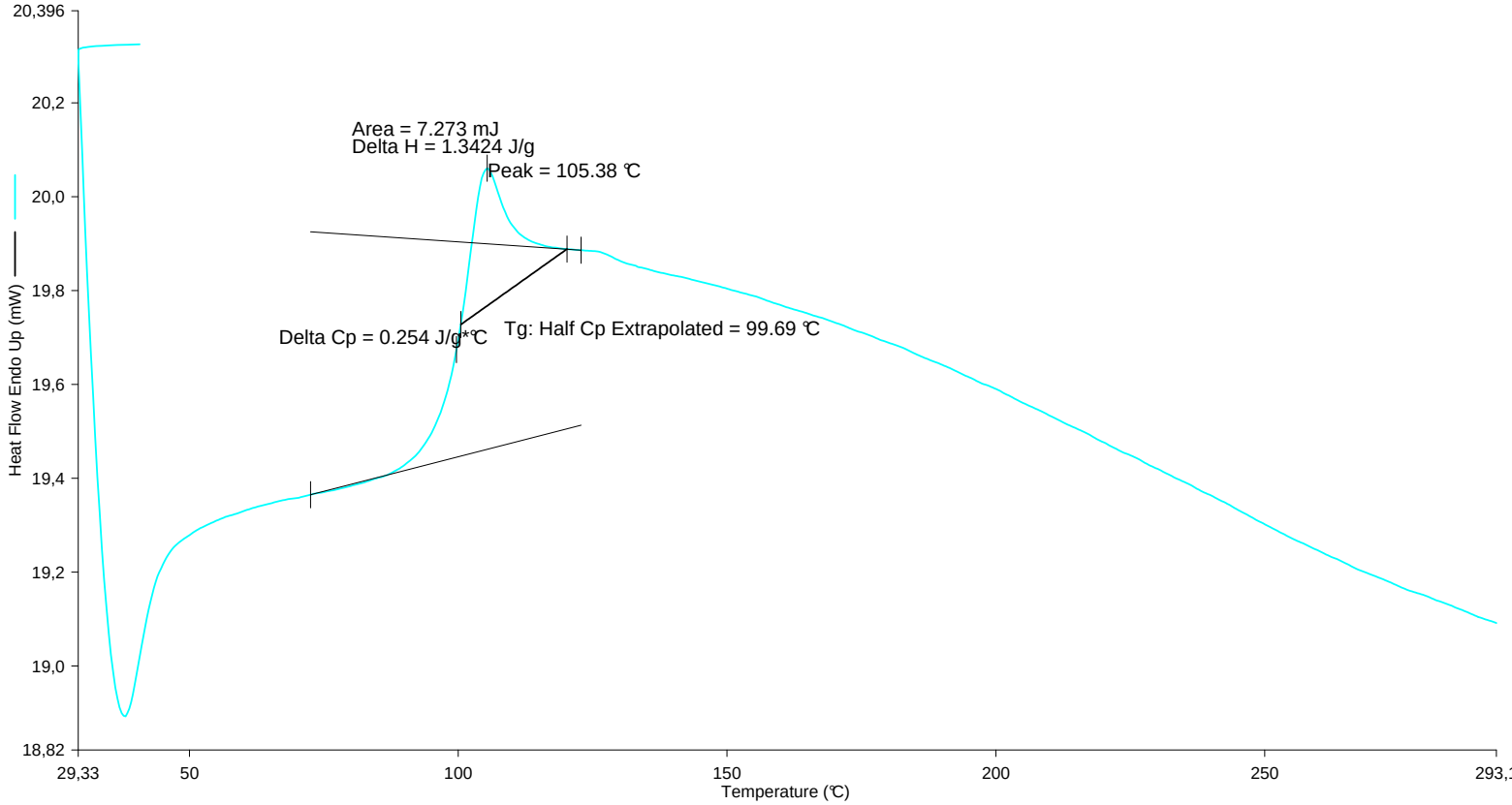
Filename:	C:\Users\DSC\Desktop\MD\G...\ps-iso-dsc.d6d
Operator ID:	MD
Sample ID:	G8-ps-iso
Sample Weight:	5.418 mg
Comment:	



10.03.2026 11:25:16

1) Heat from 30.00℃ to 300.00℃ at 20.00℃/min
2) Cool from 300.00℃ to 30.00℃ at 20.00℃/min
3) Heat from 30.00℃ to 300.00℃ at 20.00℃/min

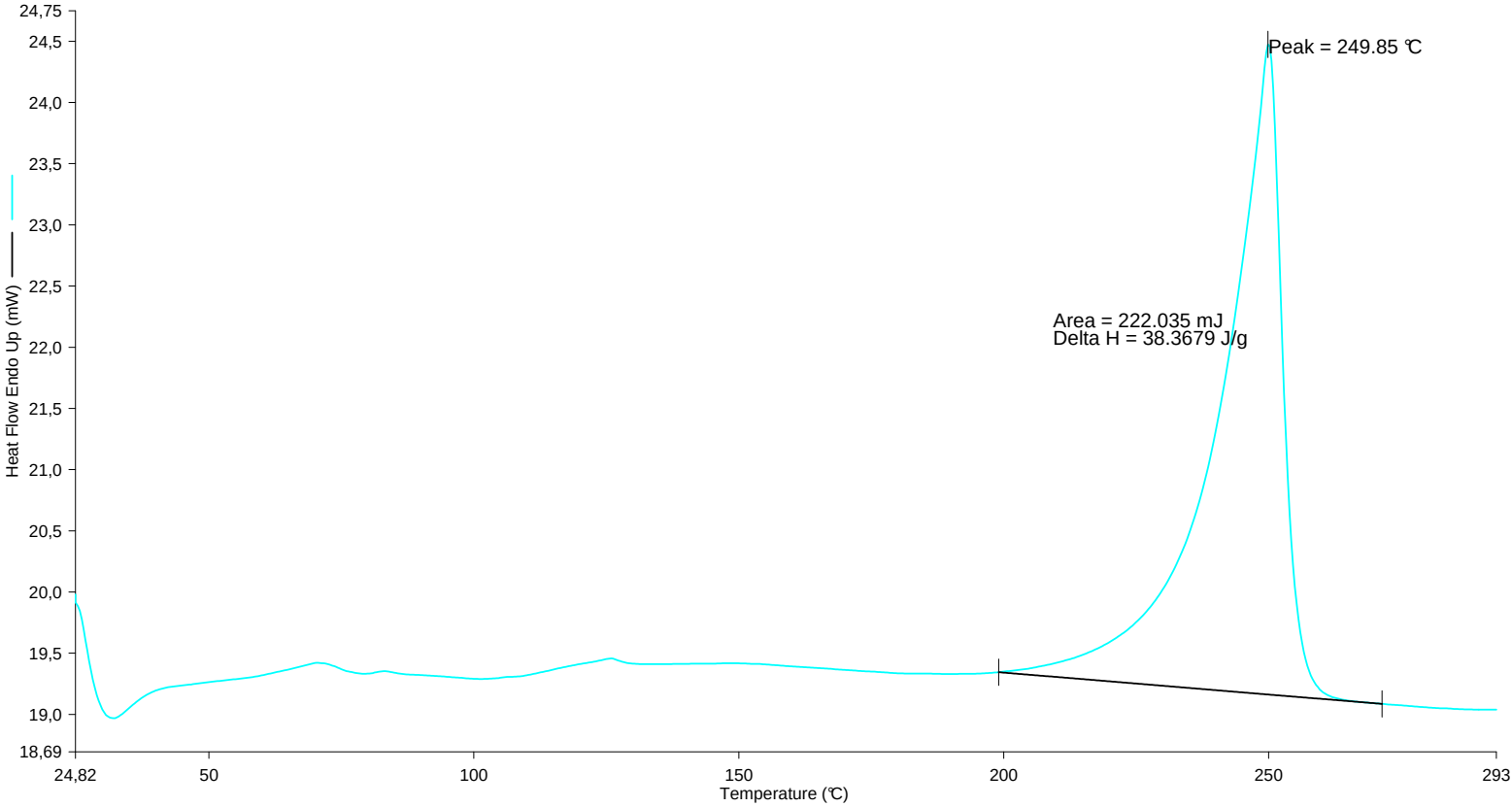
Filename:	C:\Users\DSC\Desktop\MD\G...\ps-iso-dsc.d6d
Operator ID:	MD
Sample ID:	G8-ps-iso
Sample Weight:	5.418 mg
Comment:	



10.03.2026 11:27:53

1) Heat from 30.00℃ to 300.00℃ at 20.00℃/min
2) Cool from 300.00℃ to 30.00℃ at 20.00℃/min
3) Heat from 30.00℃ to 300.00℃ at 20.00℃/min

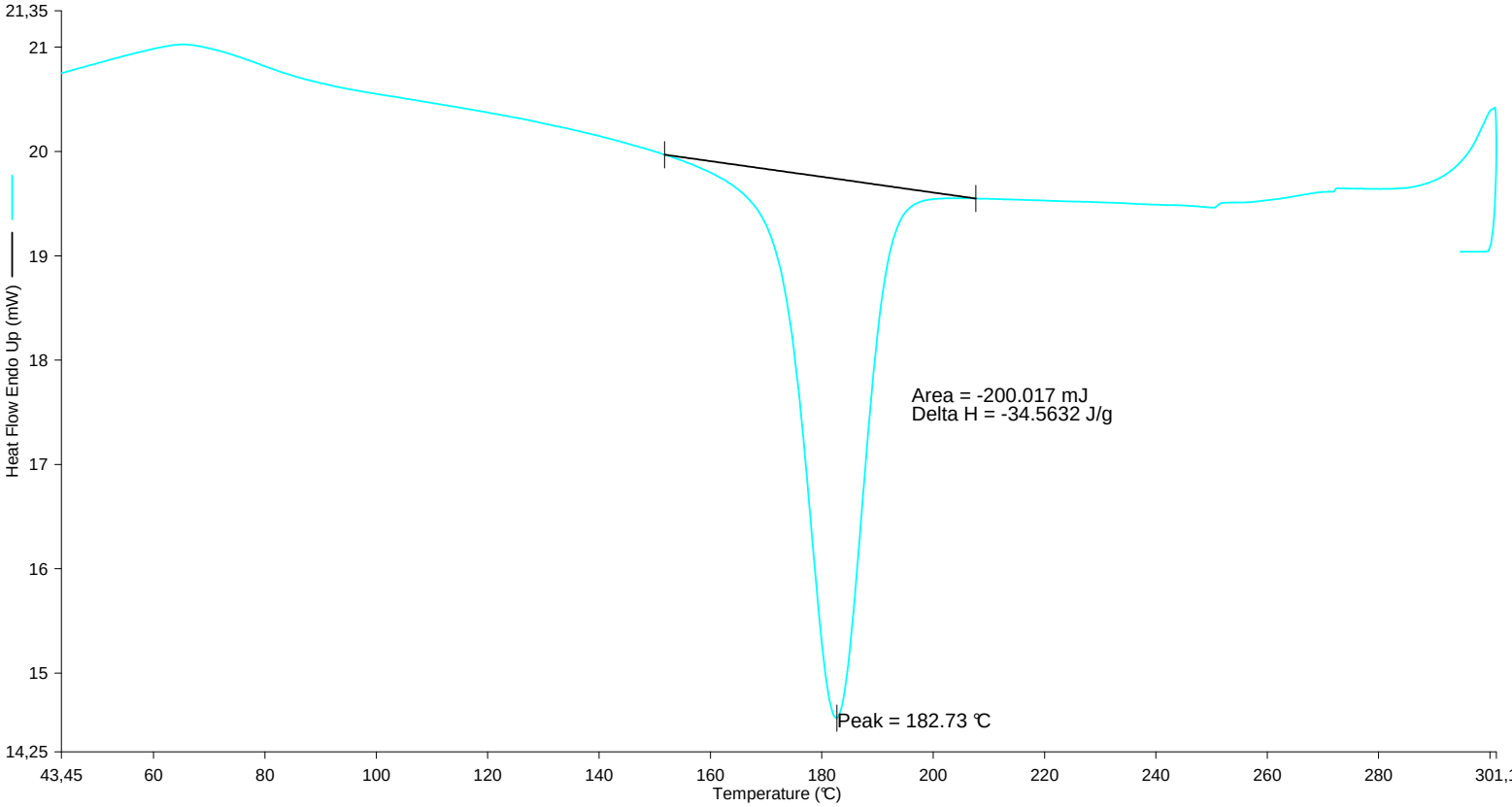
Filename:	C:\Users\DSC\Desktop\MD\G8\pet-dsc.d6d
Operator ID:	MD
Sample ID:	G8-PET
Sample Weight:	5.787 mg
Comment:	



10.03.2026 10:26:23

1) Heat from 30.00℃ to 300.00℃ at 20.00℃/min
2) Cool from 300.00℃ to 30.00℃ at 20.00℃/min
3) Heat from 30.00℃ to 300.00℃ at 20.00℃/min

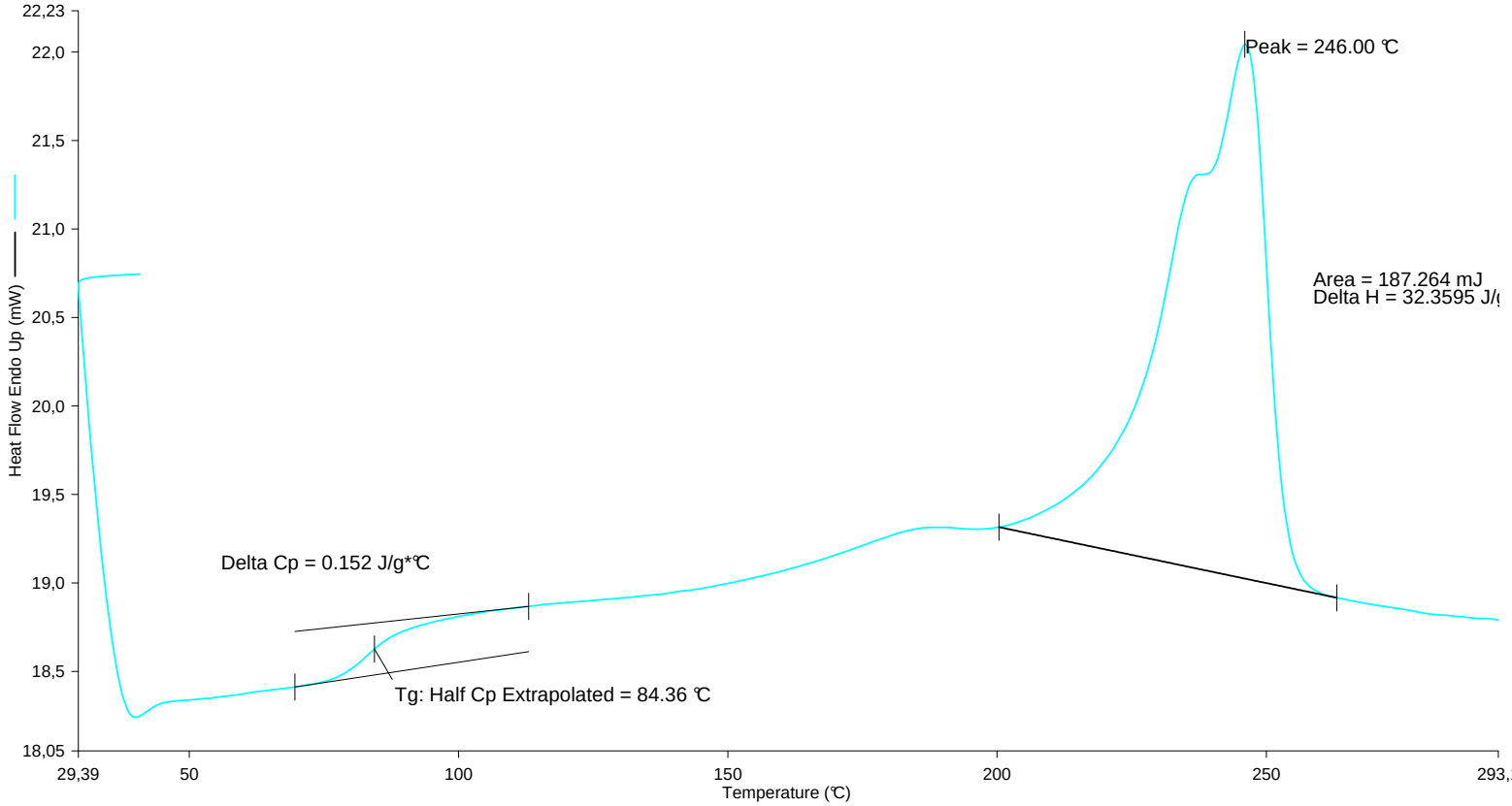
Filename:	C:\Users\DSC\Desktop\MD\G8\pet-dsc.d6d
Operator ID:	MD
Sample ID:	G8-PET
Sample Weight:	5.787 mg
Comment:	



10.03.2026 10:27:39

1) Heat from 30.00℃ to 300.00℃ at 20.00℃/min
2) Cool from 300.00℃ to 30.00℃ at 20.00℃/min
3) Heat from 30.00℃ to 300.00℃ at 20.00℃/min

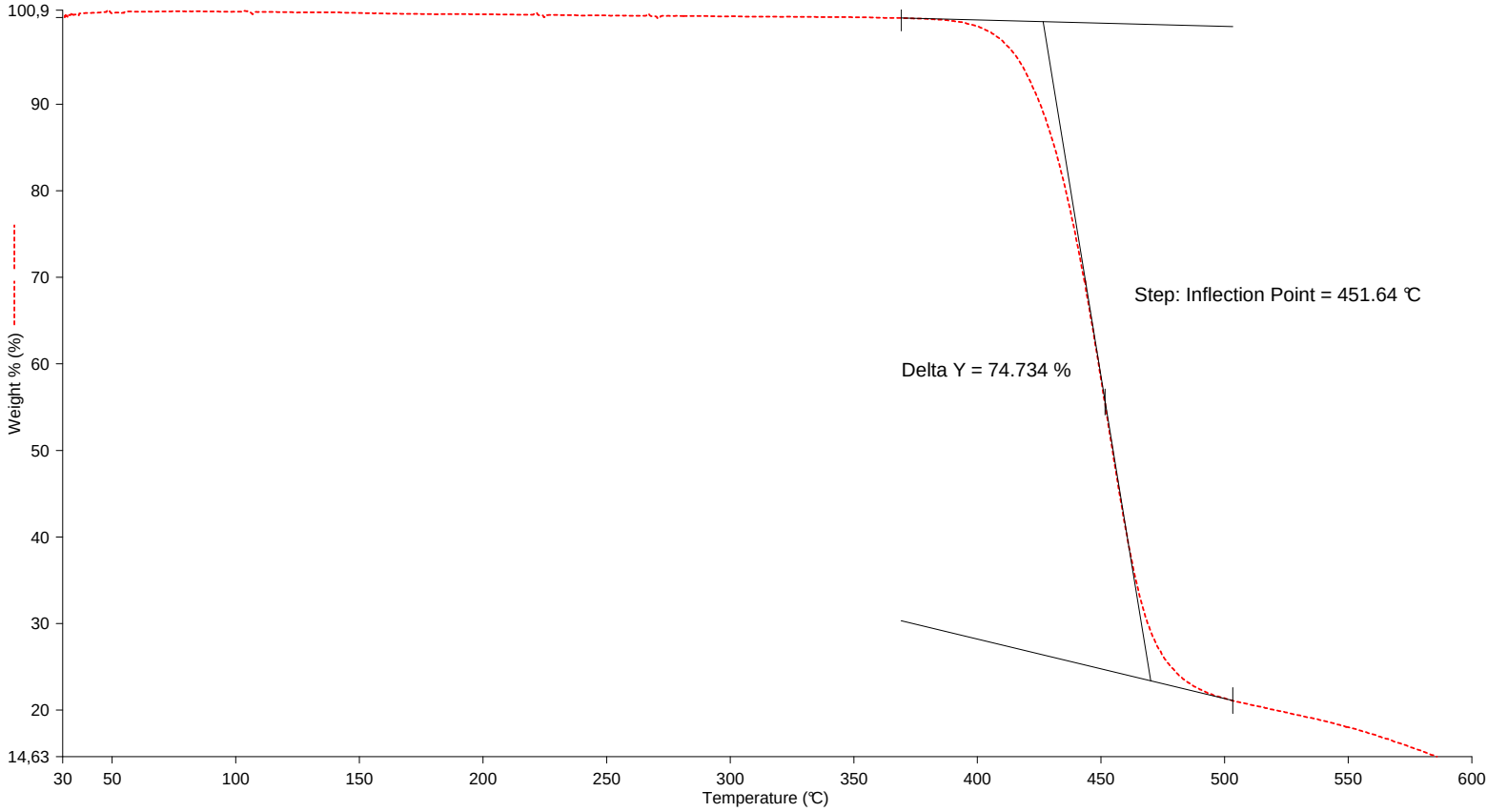
Filename:	C:\Users\DSC\Desktop\MD\G8\pet-dsc.d6d
Operator ID:	MD
Sample ID:	G8-PET
Sample Weight:	5.787 mg
Comment:	



10.03.2026 10:29:27

1) Heat from 30.00℃ to 300.00℃ at 20.00℃/min
2) Cool from 300.00℃ to 30.00℃ at 20.00℃/min
3) Heat from 30.00℃ to 300.00℃ at 20.00℃/min

Filename:	C:\Users\DSC\Desktop\MD\G...\G8-pet-tga.t6d
Operator ID:	MD
Sample ID:	G8-PET
Sample Weight:	3.597 mg
Comment:	



10.03.2026 10:42:27

1) Heat from 30.00℃ to 600.00℃ at 20.00℃/min

Literatur

[1] In *Thermal Analysis of Polymeric Materials*, (Hrsg.: B. Wunderlich), Springer, Berlin, Heidelberg, **2005**, S. 455–590.